Next Utterance Ranking Based On Context Response Similarity

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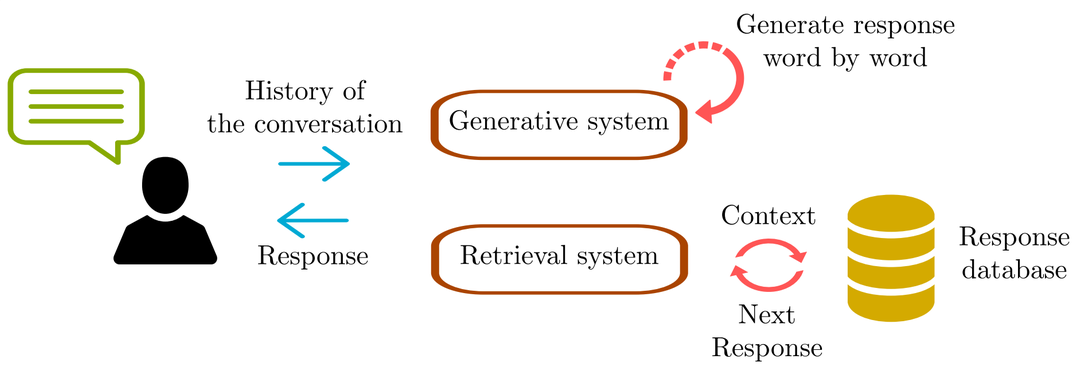
Context

- Increasing need to develop machines able to converse with humans and help them performing their daily tasks.
- Two categories of dialogue systems: generative and retrieval systems.
- *Task specific dialogue systems*: restaurant recommendation, hotel booking, etc. vs *chit-chat systems*: Cortana, Alexa, Replika, etc.
- Available data and computing power helped building data-driven dialogue systems.



Problem

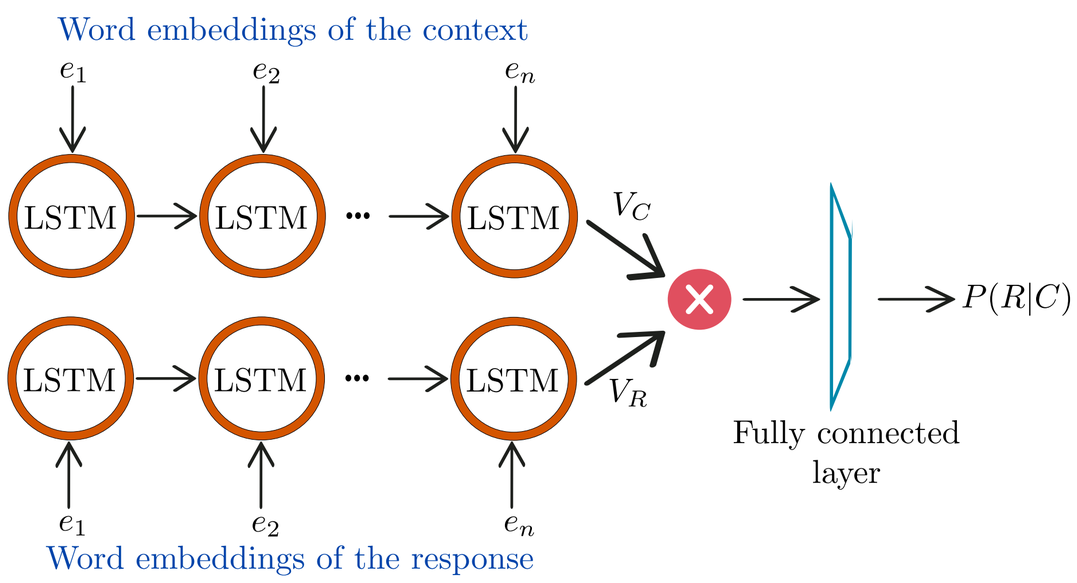
We address the problem of retrieving the next utterance of a given conversation's context and a set of candidate responses. The problem consists of:

- Understanding the context of the conversation.
- Identifying important information (words, phrases and sentences).
- Matching these information between the context and the candidate response.
- Modeling the relationship between the context and the candidate response.

	Context
u1	Hi, I can not longer access the graphical **login screen** on ubuntu 12.04
u2	what exactly happen?
u3	I can't remember the error message, would it have auto-logged to a file or should I reboot quick?
u4	you mean it won't **automaticaly start** and what happen then?
u5	it just stop at a text **screen**, but I can access the command line **login** via alt F1-6, and **start** x **manually** there. I think it might me **lightdm** that's break but I'm not sure
	Candidate responses
r1	for me **lightdm** often won't start **automatically** either. It show me console tty1 instead and I have to **start lightdm manually** ✓
r2	what about sources.list ? ✗

Model

Based on the dual encoder of [1], we propose a new response ranking system trained in an end-to-end fashion with supervised learning.



1. Represent the input words using word embeddings (initialized with Glove).
2. Word embeddings are fed in chronological order into a shared LSTM neural network.
3. The hidden layer of the network is updated every time an embedding is fed.
4. A cross product is computed between the context and the response vectors and transformed into a probability using a fully connected layer and sigmoid.
5. This classification probability is used in order to rank the candidate responses.

Experiments and results

We evaluated our system on a large public dataset: the Ubuntu Dialogue Corpus.

Dataset

# utterances (total)	7,100,000
# turns (total)	5,139,574
# words (total)	100,000,000
Min. # turns per dialogue	3
Avg. # turns per dialogue	4.94
Avg. # word per turn	10.34
# train samples	1,000,000
# test samples	18,920
# validation samples	19,560

Evaluation results

- We followed [1, 2, 3] and evaluated our approach using Recall@k metrics.
- Our utterance ranking system outperforms the state of the art systems on all metrics with a good margin.

Method	Recall@1	Recall@2	Recall@5
TF-IDF [4]	48.8 %	58.7 %	76.3 %
RNN Dual Encoder [4]	37.9 %	56.1 %	83.6 %
LSTM Dual Encoder [4]	55.2 %	72.1 %	92.4 %
BiLSTM Dual Encoder [5]	54.2 %	71.6 %	91.9 %
Similarity LSTM Dual Encoder	**62.2** %	**78.0** %	**94.9** %
Similarity BiLSTM Dual Encoder	**62.3** %	**78.2** %	**95.1** %

Error Analysis

In some cases our model was not able to retrieve the best utterance among the candidate responses. The reasons of this bad ranking may include:

- The nature of the corpus: the candidate responses were randomly sampled from the hole corpus, without any human evaluation.
- Our model fails to rank short and general responses.

	Context		Candidate responses
u1	http://www.howtogeek.com how to add screensavers to ubuntu 12.04 see also	0.99	it's only annoying when the cursor drag really slowly
	http://askubuntu.com questions how can i change or install	0.87	apt-get install hwinfo
u2	ok it won't become an issue on system upgrade	0.85	ok what is that ok just figure it out you just help me out haha
u3	then you probably just need to log out back in to restart indicator messages	**0.27**	**thank you**

Conclusion

- We proposed in this work a utterance ranking system based on dual encoder.
- Experimental results show that our approach brings significant improvements compared to the state of the art systems.
- Our new approach based on semantic and syntactic similarities between the context and response allows to better distinguish between good and bad responses.

Bibliography

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